Minimum Requirements Document

Mettler-Toledo Recrystallization Equipment Purchase
31 March 2026

Scope: AFRL/RWTE requires the purchase, installation, and training of the Mettler-Toledo EasyViewer 100 system, the ReactRaman 802L spectrometer, Raman probe, and the associated software to support recrystallization efforts.

Background: AFRL/RWTE has been directed by OUSD to develop new explosive formulations incorporating CL-20 material. Due to the characteristics and sensitivity of CL-20 material, researchers need the ability to control and analyze recrystallizations in real-time to improve safety and processing risk prior formulation. The researchers need to identify trends during the recrystallization process and make adjustments as potential issues are occurring.

Requirements/Salient Characteristics: The following Salient Characteristics are firm requirements and any "or equal" item must meet these characteristics:

- The software and equipment must integrate with the EasyMax 402 Automated Lab reactor
- Requires no manual sampling
- The Viewer shall:
    - Motorized focus for autofocus software capability
    - Resolution of 1.5 um
    - Field of view 1000 x 1000 um
    - Temperature range (-20 to 135°C) with Nitrogen purge
    - Pressure range (vacuum to 3 barg)
    - Instantaneous microscopic image collection
    - Instantaneous particle size measurements
- The software must:
    - Automatic integration into a data center which captures the experimental data from the instruments and automatically prepares this data to well-known Microsoft Office formats
    - Allows reaction parameter trends controlled by the EasyMax 402 to be displayed in real time along with crystallization trends
    - Automatic algorithm to allow for unattended adjustment of the focal point as required for in-focus images
    - Shows particle boundaries for particles considered for image analysis
    - Image analysis capabilities to determine particle size (length, width), shape (aspect ratio, equivalent diameter, circularity)
    - Highlights out of focus particles not considered for image analysis

- The spectrometer shall:
    - Shall be no larger than 325mm (L) x 206mm (W) x 142mm (H) and weigh no more than 7.5 kg
    - Include a 785 nm laser that is thermally stabilized using an integrated thermoelectric cooler, to provide a consistent and reproducible laser excitation frequency output. This is a requirement for reaction or time domain data to eliminate any shift in the position of spectral bands during an experiment due to laser temperature fluctuations that will compromise data quality and obscure chemistry information
    - Include a high-performance CCD detector that is thermoelectrically cooled and operates at a temperature not greater than -40ºC.
    - provide a spectral range with a starting point of not greater than 150 cm^{-1} Raman shift and an ending point of not less than 3400 cm^{-1} Raman shift.
- The probe shall:
    - Capable of delivering, through software control, at least 375 mW of laser power at the probe tip.
    - Capable of achieving a probe temperature range of -40ºC to 300ºC
    - An interchangeable probe with a 305 mm wetted length for use in 50-1000mL. This probe will fully integrate with Mettler Toledo's EasyMax 402 Automated Lab Reactor owned by the Eglin AFB.
- The supporting software package shall be designed specifically for in-situ, real-time reaction analysis applications that include the following elements:
    - Guided setup ensuring high-quality data collection.
    - Laser power shall be easily adjusted via incremental settings in the software.
    - Exposure time have both a manual or automatic selectable/changeable setting.
    - Calibration to a cyclohexane standard meeting ASTM E1840 specifications ensures optimal data quality
    - Integrated cosmic ray removal ensures best quality spectra spectral results
    - Instrument profile normalization allowing system to system data reproducibility
    - Offline data from HPLC or other analytical methods can be used to fit trends; converting them from qualitative to quantitative trend measurements in real-time from beginning to end of a reaction
    - Automatically profile reaction spectral data to represent a fully automated set of reaction chemistry behavior
    - Tool to quickly analyze and profile a reaction
    - Active Baseline Correction to eliminate non-chemistry related events and highlights only the changes due to the reaction – dynamically removing non-linear and relatively intense baseline artifacts that otherwise complicate and impede data analysis
    - Automatically rejects erroneous and confusing spectral spikes

    - Interactive 3D waterfall reaction view – pan, zoom, and profile directly from the 3D waterfall plot
    - User defined trends (e.g., addition, subtraction, multiplication, etc.) for easy reaction profiling for more advanced trend analyses
- Availability of a one year maintenance support package for the viewer, software, and spectrometer